

CALCULUS AND DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: 75 | Total Questions: 3 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	Total
Marks	25	25	25	75

Question 1

[25 Marks]

Evaluate the definite integral $\int_0^1 x^2 e^x \, dx$ using integration by parts. Show all steps and simplify the final answer.

Question 2

[25 Marks]

Solve the first-order linear differential equation $(dy / dx) + 2y = 4e^{-x}$ with the initial condition $y(0)=1$. Provide the explicit solution $y(x)$.

Question 3**[25 Marks]**

Consider the scalar field $u(x,y) = x^2 y - y^3$. (a) Compute the gradient ∇u at the point $(1,-1)$. (b) Find the directional derivative of u at $(1,-1)$ in the direction of the vector $\mathbf{v} = (3,4)$. (c) Sketch the surface $z = u(x,y)$ over the region $-2 \leq x \leq 2, -2 \leq y \leq 2$.
